Infection and Classification

Leprosy is highly infectious, but the attack rate is low. The major reason for this low attack rate is that most people are genetically unable to provide the mycobacteria with what they need to survive, because they lack the type of genes the bacterium requires to reprogram the host cell for its survival and multiplication. Cell-mediated immunity (CMI) plays a crucial role in individuals who develop leprosy.

To predict complications and stratify patients according to CMI, the **Ridley-Jopling classification** (Figure 4) is widely used. This spectrum includes:

- **Polar tuberculoid (TT)**: Characterized by a single, well-defined skin lesion or an enlarged nerve. Bacilli are not detectable, and there is strong CMI against *M. leprae* antigenic determinants.
- **Polar lepromatous (LL)**: Presents with nodules and/or plaques, symmetrically enlarged nerves, or diffuse skin infiltration (Lepra bonita). Numerous bacilli are present, but there is no detectable CMI against *M. leprae*.
- **Borderline group**: Includes the majority of patients and is subdivided into: - **Borderline tuberculoid (BT)**: Lesions resemble TT but are less well-defined. - **Borderline lepromatous (BL)**: Features more bacilli and less granulomatous inflammation than BT, approaching LL. - **Mid-borderline (BB)**: Characterized by "punched-out" (centrally clear) or dome-shaped lesions.

This classification helps guide prognosis, treatment, and monitoring for reactions.

Diagnostic Approaches in Leprosy

The **Ziehl-Neelsen stain** using 3% hydrochloric acid may decolorize *M. leprae*, potentially rendering smears falsely negative. Alternative methods such as **PCR** or **NASBA** can detect bacilli, though these are often negative in **paucibacillary (PB)** patients, similar to smears. However, smear microscopy, immunological, and molecular techniques are valuable in diagnosing **multibacillary (MB)** leprosy, monitoring treatment response, and detecting relapses.

Other laboratory tools include:

- **Anti-PGL-1 antibody titre**: PGL-1 is a species-specific glycolipid in the *M. leprae* cell wall. Elevated titres are useful in MB leprosy but may be positive in asymptomatic contacts and negative in PB cases. It aids in classifying PB vs. MB leprosy and monitoring treatment and relapse.
- **LID-1 (synthetic antigen)**: Recently introduced, but adds little diagnostic value.
- **Lymphocyte transformation tests or cytokine assays (e.g., IFN-γ, IL-2)**: Have shown limited utility for diagnosis to date.
- **Lepromin test (Mitsuda)**: An older test that is positive in TT and BT (strong CMI) and negative in LL (absent CMI). However, it can yield variable results in healthy individuals and is not diagnostic. Due to its biological origin, it carries a theoretical risk of sensitization, leading to opposition against its use. It remains useful primarily for classification.

Histopathology and Biopsy

Histopathology is valuable for diagnosis, classification, and identifying leprosy reactions. Immunopathology shows promise but remains experimental.

Key considerations for biopsy: - In **tuberculoid leprosy**, take biopsy from the **edge (rim)** of the lesion. - In **lepromatous leprosy**, biopsy the **center** of the lesion. - Use **Fite-Faraco staining** for optimal visualization of *M. leprae* (more sensitive than Ziehl-Neelsen due to lower acid concentration).

For **pure neural leprosy**: - Perform nerve biopsy from a **small cutaneous or subcutaneous nerve**. - From larger nerves, **fine-needle aspiration (FNA)** can be used for cytology, bacteriology, and PCR.

A challenge is that histopathological findings can vary even within a single biopsy specimen.